PSY511

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Lesson 1:
Introduction to Environmental Psychology

Origins of Environmental Psychology

Environmental psychology began in the 1960s, with researchers from architecture, sociology, and psychology studying how environments affect human behavior and well-being. Initially focused on built environments like homes and offices (architectural psychology), it later included natural environments like parks and forests (conservation psychology).

Topics in Environmental Psychology

Environmental psychology covers various topics, including:
- Environmental perception, attitudes, and values
- Environmental behavior and decision-making
- Environmental stress and coping
- Environmental restoration and well-being
- Environmental education and communication

Impact of Environments on Feelings and Behavior

Different environments evoke different feelings and behaviors:
1. In a comfortable room, you may feel pleasant and motivated.

2. In a traffic jam, you may feel stressed and irritable.

Modern Environmental Challenges
Rapid industrialization has led to complex environmental problems like acid rain, unreliable nuclear power, and toxic waste. Chemicals once confined to labs are now common in households, and both rich and poor contribute to environmental damage.

The Environmental Debate
There are two main perspectives on environmental issues:

The Pessimists: Believe the environment is so damaged it will be unlivable by 2100, pointing to the loss of species, deforestation, and pollution.
The Optimists: Highlight achievements in agriculture, medicine, and technology, believing human creativity can overcome environmental challenges.

Unresolved Debate
The debate continues, with differing views on whether technology or behavioral change can solve environmental problems. The 1992 UN Earth Summit highlighted these concerns.

Definition of Environmental Psychology
Environmental psychology is the study of the interactions between humans and their environments. It includes various disciplines and aims to understand and improve these interactions.

Goals of Environmental Psychology
- Understand how environments impact behavior, emotions, and cognition
- Improve environment design to promote well-being and sustainability
- Foster environmental awareness and conservation

Objectives of the Course
1. Learn the history, scope, and methods of environmental psychology
2. Understand the interaction between human behavior and the environment
3. Explore its interdisciplinary nature and connections with other fields
4. Apply environmental perspectives to various issues
5. Examine theoretical and empirical developments

6. Anticipate future trends and challenges
7. Analyze environmental challenges in Pakistan

Common Assumptions of Environmental Psychology

1. Earth is our only suitable habitat.
2. Earth's resources are limited.
3. Life profoundly affects Earth.
4. Human land use has cumulative effects.
5. Sustained life depends on ecosystems, not just individual organisms or populations.

We will discuss these assumptions in detail in the next lesson.

LESSON 2

COMMON ASSUMPTIONS OF ENVIRONMENTAL PSYCHOLOGY

The term "environment" can be interpreted differently by various disciplines. Despite this, there are some common assumptions in environmental science:

1**. The Earth is the Only Suitable Habitat We Have:**

- The Earth is our only home, and its resources are limited. Human activities impact the planet significantly, and our survival depends on adapting to these constraints and opportunities sustainably.

2. The Earth as a Planet Has Been Profoundly Affected by Life

- Human activities change the Earth's landscape and atmosphere. These changes affect climate and weather, highlighting the need for better understanding and responsible behavior.

3.The Outcomes of Land Use by Humans Tend to Be Cumulative

- Human activities have long-term and sometimes irreversible effects on the environment. We have a responsibility to minimize negative impacts for ourselves and future generations.

4. Sustained Life on Earth is a Characteristic of Ecosystems, Not Individual Organisms or Populations:

- No single species can sustain itself without others. Humans are part of a complex ecosystem, and understanding this interdependency is crucial for sustainability.

Environmental psychology integrates these assumptions to understand the relationship between humans and their environment.

Basic and Applied Behavioral Science

Basic Behavioral Science:

- Studies human behavior to discover general laws and theories. Example: Behavioral psychology.

Applied Behavioral Science: Uses scientific knowledge to solve practical problems. Example: Applied Behavior Analysis (ABA).

Current Events Influencing Environmental Psychology

1. Population Trends:

- The world population is projected to reach 10 billion by 2030. Some regions, like Sub-Saharan Africa and parts of Asia, have already exceeded their local carrying capacity, leading to environmental degradation.

Population Statistics in Pakistan:

- Population (mid-2007): 169.3 million
- Births per 1,000 population: 31
- Deaths per 1,000 population: 8
- Rate of natural increase: 2.3%
- Projected Population (2025): 228.8 million
- Projected Population (2050): 295 million

The dramatic increase in world population is a key factor in the development of Environmental Psychology.

In this lecture, only “Population Trends” was discussed in detail. The remaining current events will be discussed in the next lecture.

Lesson 3: Theories in Environmental Psychology

Current Events Influencing Environmental Psychology

Resource Depletion and Environmental Degradation

- **Erosion of Agricultural Soil**: Increasingly common, affecting productivity.
- **Salinization:** Salty soil affects irrigated farmland.
- **Lake Acidification**: More prevalent.
- **Tropical Forest Loss**: Significant in the Amazon Basin.
- **Desert Expansion**: World's deserts cover 800 million hectares and are growing.
- **Water Shortages**: Expected to worsen with increased withdrawals.
- **Conflict Over Water**: Shared rivers (e.g., Euphrates, Ganges)

Public Policy and the Environment

- **Policy Changes:** Efforts in reforestation, detoxification, soil management, energy conservation, and recycling.
- **Scientific Evidence**: Required for policy decisions, fostering the growth of environmental psychology.
- **Economic Considerations**: Policies need to provide alternatives to industries affected by environmental changes, like replacing plastic bag production.

Human Behavior and the Environment

- **Complex Solutions**: Necessary to address global issues like poverty, degradation, and conflict.
- **Role of Psychologists:** Focus on understanding and changing human behavior to address environmental problems.
- **Behavior Modification**: Examples include token economies to promote eco-friendly habits.

Theories in Environmental Psychology

Historical Influences

- **Broad in Scope:** Theories that address numerous phenomena.
- **Focused in Scope**: Theories that target specific issues.
- **Empirical**: Data-driven and scientific.

- **Non-Empirical**: Based on logic and intuition.

Key Theories

1. **Geographical Determinism:**
   - Civilizations rise or fall based on environmental challenges.
   - Intermediate environmental challenges enhance development.

2. Ecological Biology:
- Mutualism between organisms and their environment.
- Changes in one element affect the whole system.

3. Behaviorism (Interactionism:
- Human behavior is influenced by environmental context and personal variables.
- **Example**: Muhammad Ali Jinnah's life influenced by both personal characteristics and environmental factors.

4. Gestalt Psychology: - Focuses on perception and cognition.
- Perception of stimuli, not just stimuli themselves, determines behavior.

Mini-Theories

- **Arousal Theories**: Performance peaks at intermediate arousal levels (Yerkes-Dodson law).
- **Stimulus Load Approach:** Analyzes the impact of sensory input on behavior.
- -**Behavioral Constraint Approach**: Focuses on how restrictions affect behavior.
- **Adaptation Level Approach**: Looks at how people adjust to environmental changes.
- **Environmental Stress Approach:** Examines the impact of stressors on behavior.
- **Ecological Approach:** Considers group behavior and environmental interactions.

Arousal Theories

- Performance and Arousal: Performance peaks at moderate arousal but drops at very high or low arousal.
- **Factors Impacting Arousal:**

- **Temperature**: High temperatures increase arousal, potentially harming performance.
- **Personal Space:** Optimal space can enhance performance.
- **Noise Levels:** Moderate noise can boost performance, while excessive noise can harm it.
- **Physiological Responses**: Changes in heart rate

Lesson 3: Theories in Environmental Psychology

Resource Depletion and Environmental Degradation

Resource depletion and environmental degradation are critical issues globally. Problems include soil erosion, salinization, and lake acidification. Tropical forests in the Amazon and desert areas are declining. Water shortages and pollution are expected to worsen, with significant potential for conflict over shared water resources. Industrialization, urbanization, and population growth exacerbate these problems, highlighting the need for environmental psychology to address these issues.

Public Policy and the Environment

Policy changes and actions by governments, businesses, and individuals are essential for preserving the earth's carrying capacity. Examples include reforestation, detoxifying chemical by-products, soil management, energy and material conservation, and family planning. Public policy relies on scientific evidence, driving the need for environmental psychology to inform these policies and encourage pro-environmental behaviors.

Human Behavior and the Environment

Human behavior significantly impacts environmental issues. Psychologists focus on behavior modification to promote eco-friendly habits, such as using a token economy to reward recycling or energy-saving behaviors. Understanding human behavior and its influence on the environment is crucial for developing effective solutions.

Theories in Environmental Psychology: Environmental psychology is interdisciplinary, incorporating theories from various disciplines. These theories can be broad or focused, empirical or non-empirical.

Historical Influences

- **Geographical Determinism:** This theory suggests that environmental challenges shape civilizations. Moderate challenges promote development, while extreme conditions can be detrimental.
- **Ecological Biology:** This approach emphasizes the interdependence between organisms and their environment, forming a holistic system where changes in one element affect the whole system.
- **Behaviorism:** This perspective considers both environmental context and personal variables to predict behavior accurately.
- **Gestalt Psychology**: Focuses on perception and cognition, suggesting that behavior is determined by how stimuli are perceived rather than the stimuli themselves.

Arousal Theories in Detail

-**Performance and Arousal:** Performance is optimal at intermediate arousal levels but decreases with too much or too little arousal.

- **Physiological Responses:** Changes in the environment can lead to physiological responses such as changes in heart rate, blood pressure, and adrenaline secretion.

- **Environmental Factors Impacting Arousal:** Factors like temperature, personal space invasion, and noise levels can influence arousal and performance.

Understanding these theories helps us address environmental issues by considering both human behavior and the broader environmental context.

Lesson 4: Arousal Theories

Behaviorism (Behaviorist Perspective or Interactionism)

Behaviorism comes from psychology and critiques personality theories' inability to fully explain human behavior. It suggests that both environmental context and personal factors (like personality and attitudes) better predict behavior. Humans interact with their environment similarly to other species. For instance, the relationship between humans and elephants spans cooperation to conflict. An individual's attitudes heavily influence their behavior, such as choosing to use harmful plastic bags due to personal preference. William James emphasized that changing one's perspective can alter their life.

Gestalt Psychology:

- **<u>Developed in Germany</u>** alongside behaviorism, Gestalt psychology focuses on perception and cognition. It suggests we perceive objects, people, and settings as whole entities, not just the sum of their parts. For example, a child might aspire to be a pilot based on processed information about pilots, while an employer may reject a candidate due to a negative perception. Behavior is influenced more by how we perceive stimuli than by the stimuli themselves.

<u>Mini-Theories</u>

- No single theory fully explains the environment-behavior relationship due to a lack of comprehensive data, varied relationships, inconsistent methods, and incompatible variable measurements. Instead, several “mini-theories” help understand specific organism-environment interactions.

  Arousal Approach: Focuses on arousal’s effect on performance.

  Stimulus Load Approach: Examines the impact of environmental stimuli on individuals.

  Behavioral Constraint Approach: Looks at how constraints in the environment affect behavior.

  Adaptation Level Approach: Studies individual differences in response to environmental changes.

  Environmental Stress Approach: Investigates stress responses to environmental factors.

  Ecological Approach: Analyzes group behavior in environmental contexts.

Arousal Theories

- Arousal theories examine how arousal impacts performance, typically showing an inverted-U relationship where performance peaks at moderate arousal levels and drops with too high or low arousal. This relationship is known as the Yerkes-Dodson law. For instance, performance may vary with temperature changes: moderate heat boosts performance, while extreme heat reduces it.

<u>Factors Impacting Arousal Level</u>

- **Temperature**: High temperatures increase arousal, potentially harming performance.
- **Personal Space:** Optimal space can enhance performance.

- **Noise Levels:** Moderate noise can boost performance, while excessive noise can harm it.
- **Physiological Responses**: Changes in heart rate
- Environmental changes can alter physiological responses such as heart rate, blood pressure, and respiration. For example, high temperatures cause blood vessel dilation and increased heart rate. Extreme conditions like scarcity of food or water can lead to lower blood pressure and insufficient brain oxygen.

Arousal and Nervous System

- Arousal is linked to activity in the brain's reticular activating system. Consistencies and breaks in environmental patterns also affect arousal. Unusual events, like seeing a person crying in a park, can heighten arousal and impact behavior. People tend to prefer moderate arousal levels, often designing environments to maintain this balance for optimal performance.

Lesson 5: Stimulus Load, Behavioral Constraint, and Adaptation Level Theories

Stimulus Load Theories

Stimulus load theories suggest that humans have a finite capacity to process information. When the environmental inputs exceed this capacity, people tend to filter out less critical information and focus on the most relevant stimuli. This selective attention allows individuals to manage the demands of their environment effectively, though it can be a limiting factor when those demands are excessively high.

Example:

- When driving during rush-hour traffic, drivers concentrate on the vehicles and road signs around them, while tuning out less relevant stimuli like radio commentary or the view of the sky. This focus enhances driving performance by prioritizing safety-related stimuli. However, if a driver ignores critical road signs due to overwhelming traffic conditions, their performance can suffer, potentially leading to navigational errors.

- Under excessive environmental demands, individuals might only focus on the most critical information, filtering out everything else. This can lead to errors in judgment and decreased tolerance for frustration. For instance, an exhausted driver might not notice a traffic light change, leading to dangerous driving behavior.

Real-World Application:
An athlete describes how they narrow their focus during a race: "I concentrate on the track, the race, the blocks, and what I have to do. The crowd fades away, and the other athletes disappear. It's just me and this one lane."

Stimulus load theories also explain behaviors in stimulus-deprived environments, such as submarines or prisons, where understimulation can be as distressing as overstimulation. Cabin fever is an example of understimulation leading to behavioral issues.

Dimensions Contributing to Stimulus Load:

1. **Intensity**: High-volume music versus low-volume music.
2. **Novelty**: Unique TV channels grabbing attention.
3. **Complexity**: Navigating rush-hour traffic.
4. **Temporal Variation:** Changes in noise levels, like an AC unit's sound suddenly stopping.
5. **Surprisingness**: A restaurant made of ice and chocolate.
6. **Incongruity**: A baby sitting with a lion.

Stimulus to Behavior Relationship:

- Stimulus load theories imply that by altering environmental stimuli, human behavior can be influenced. For example, designing schools and offices to foster positive and productive behaviors.

  Behavioral Constraint Theories

  - Behavioral constraint theories focus on the limitations imposed on individuals by their environment, whether these are real or perceived. For instance, a student might feel anxious on a train ride, perceiving danger and attempting to regain control through strategies like making friends or reading a travel guide.

- **Psychological Reactance:**

- When people feel they have lost control over their environment, they experience discomfort and attempt to reassert control. If they perceive control over their environment, even if it's an illusion, it can mitigate negative outcomes. For example, nursing home residents given control over their environment displayed better mood and activity levels.

Learned Helplessness:

- Repeated failure to regain control can lead to learned helplessness, where individuals feel their actions no longer impact their environment, potentially leading to depression.

Adaptation Level Theories

- Adaptation level theories propose that optimal behavior occurs at an intermediate level of stimulation. Both excessive and insufficient stimulation can negatively affect emotions and behaviors.

Example:

- A highly stimulating home interior might cause restlessness, while a dull one might lead to depression. An optimal environment balances stimulation to create comfort.

Adaptation and Adjustment:

- **Adaptation**: Changing response to the environment, like a political prisoner focusing on reading.
- **Adjustment**: Changing the environment to meet needs, like turning up the thermostat in response to cold.

Personal Adaptation Levels:

- Individual preferences for excitement explain why people react differently to the same situation. For example, someone enjoying lively parties versus another preferring quiet settings.

Lesson 6

Ecological Theories and Comparison of Theories

Environmental Stress Theories

Stress theories emphasize the role of physiological, emotional, and cognitive responses in the interaction between organisms and their environment. Stress responses occur when environmental features exceed optimal levels, causing a disturbance in homeostasis.

Types of Stress:

1. **Eustress**: Positive stress that motivates, like preparing for a debate.
2. **Distress**: Negative stress that can lead to failure, like experiencing a panic attack.

Stages of Stress Response (Hans Selye):

1. **Alarm Reaction:** Initial acknowledgment of stress, preparing the body to deal with it.
2. **Resistance**: Strategy to cope with stress, consuming more energy.
3. **Exhaustion**: Long-term stress leads to resource depletion and survival mode.

Indicators of Stress:

1. **Physiological**: Changes in blood pressure, heart rate, and blood sugar levels.
2. **Emotional**: Feelings of irritability, anxiety, and depression.

Ecological Theories

Ecological theories emphasize the fit between organisms and their environment. For example, a person with analytical skills fitting well in a banking career. Behavior settings, like schools or offices, are evaluated based on how well they support the intended behaviors.

Example:

A classroom's standing pattern of behavior includes lecturing, listening, and taking notes, with physical elements like a chalkboard and chairs supporting these behaviors.

Cultural Purpose: The interdependency between behavior and physical settings defines cultural purposes. Knowing about the setting helps predict behaviors within it. For example, students' behaviors change dramatically when moving from a classroom to another environment.

<u>**Comparison of Theories:**</u>

- **Stimulus Load Theories:** Focus on managing information capacity and its impact on behavior.
- **Behavioral Constraint Theories**: Emphasize environmental limitations and the perception of control.
- **Adaptation Level Theories**: Highlight optimal levels of stimulation for balanced behavior.
- **Ecological Theories**: Stress the fit between environment and behavior, **focusing on behavior settings.**

Understanding these theories helps in designing environments that optimize human behavior, enhancing productivity, well-being, and adaptability.

Lesson 7
Environmental Psycholo

Ecological Theories

****Undermanning (Understaffing):****
- Too few people for available roles/resources.
- Individuals must take on multiple roles.
- Example: Small high school where students participate in multiple activities.

Overmanning (Overstaffing):
-Too many people for available resources.
Leads to overcrowding and resource strain.
Examples: Crowded pools, commuter trains, Super Bowl crowds.

<u>**Comparison of Theories**</u>
General vs. Specific Scope
- General Scope: Broad range of phenomena (e.g., Arousal, Stress, Ecological Approach).
- **-Specific Scope:** Focus on specific phenomena (e.g., Stimulus Load, Adaptation Level, Behavioral Constraint).

Key Statements and Predictions

Arousal Approach:

Predicts changes in behavior based on physiological and psychological arousal.

Stimulus Load Approach:

Predicts social and behavioral consequences of over/under-stimulation.

Adaptation Level Approach:

Predicts optimal behavior at intermediate stimulation levels.

Stress Approach:

Predicts coping behavior under stress.

Behavioral Constraint Approach:

Predicts responses to loss of control or threats.

Ecological Approach:

Describes how behavior settings impact behavior

Real-Life Application: Amir

Amir moves to a busy city for university, experiences stress and health issues, and eventually returns to his village.

Questions to consider: Amir's stress levels, environmental influences, and interactions between his personality and the environment.

The Present Framework and Future Directions

Purpose:

- Understand organism-environment relationships.

Origin:

- Derived from psychology and environmental psychology.

Target:

Unify theories under one model.

Specific Focus:

How environmental factors affect emotions and behavior.

Seven Postulates

1. Environment-Behavior Reciprocity:

Physical and social environments interact with behavior.

2. Steady-State Influence:

Environments exert a constant influence until disrupted.

3.Indirect Influence via Emotions:
Environment affects emotions, which then influence behavior.
4. Three Dimensions of Emotions:
Pleasure–displeasure, degree of arousal, dominance–submissiveness.
5**. Behavior Depends on Emotions**:
Behavior influenced by the extent of emotions aroused.
6. Goal-Directed Behavior:
Behavior aims to achieve a steady state.
7. Pathological Environment:
When goal-directed behavior fails, the environment disrupts functioning.
Detailed Explanation of First Postulate
1A. Reciprocal Relationship:Continuous interaction between behavior and environment.
Example: Classroom setup affects behaviors, which in turn change the environment.
1B. Measurement of Characteristics:
Precise measurement helps understand environmental effects.
Example: Determining the exact temperature that affects learning.
Second Postulate
2A. Steady-State Influence:
Environment influences behavior below awareness.
Example: Desk arrangements affecting student interactions.
2**B. Disruption from Dramatic Changes:**
Significant changes require conscious adaptation.
Example: Temperature changes requiring adjustments.
2C. Disruptions from Expected Levels:**
Deviations from expected conditions disrupt steady-state influence.
Example: Different study needs for different students.

Next Lecture
Discussion of the third postulate in detail.

Lesson 8
The Present Framework of Environmental Psychology (II)

Third Postulate: Indirect Influence of Environment

Concept: The environment influences people's emotions, which in turn affect their behavior.

Examples:

- Temperature, crowding, noise, air pollution, and radio news can impact emotions and behaviors.
- Annoying sounds might cause irritability, while pleasant scents can lead to positive interactions.

Fourth Postulate: Three Dimensions of Environment-Evoked Emotions

- Dimensions:
  - **Pleasure-Displeasure**: Emotional response to an event (e.g., winning a lottery brings pleasure, losing a phone brings displeasure).
  - **Degree of Arousal**: Physical reactions related to emotions (e.g., adrenaline rush from excitement).
  - **Dominance-Submissiveness:** Sense of control over the environment (e.g., feeling powerful after success or powerless after failure).

Fifth Postulate: Behavior Depends on Emotional Dimensions

-**Concept**: Emotional dimensions (pleasure, arousal, dominance) influence behavior.

- **Examples**:
  - Anxiety (displeasure, high arousal, submissiveness) might lead to avoidance.
  - Anger (displeasure, high arousal, dominance) might lead to confrontation.

Sixth Postulate: Goal-Directed Behavior

- **Concept**: People try to maintain a steady state by adjusting their environment or behavior.
- **Examples**:
  - Moving to a cooler place when it's too hot.
  - Changing tasks to make them more enjoyable.

Seventh Postulate: Pathological Environment

- **Concept**: When maintaining a steady state requires too much effort, the environment is considered pathological.
- **Examples**:
  - Prolonged noise exposure can be harmful.
  - High population density can affect social behavior negatively.

Model Interaction

- **Factors**: Noise, pollution, weather, architecture, and social situations influence behavior.
- **Personal Variables**: Adaptation levels, expectations, goals, moods, and sentiments mediate the impact of the environment.
- **Outcome**: The environment either helps achieve a steady state or causes behavioral disruptions.

Lesson 9
Elementary Psychophysics/Perception and its Cognitive Bases

Elementary Psychophysics

Elementary psychophysics studies how humans detect physical stimuli.

Objectives of Elementary Psychophysics:

1. **Stimulus Detection:**
   - Understanding how we detect changes in energy like light, sound, and temperature.
   - Absolute threshold: The minimum intensity required for a stimulus to be detected.

2. **Stimulus Recognition:**
   - Recognizing what a detected stimulus is by matching it with previous experiences.
   - Context and available alternatives affect recognition.

3. **Stimulus Discrimination:**
   - Differentiating between similar stimuli.

- **<u>Just Noticeable Difference (JND):</u>** The smallest change in stimulus intensity that can be detected.
- **Weber's Law**: High-intensity stimuli require larger changes to be noticeable.

4. **Stimulus Scaling:**
   - Determining the magnitude of a stimulus.
   - Scaling helps us decide how to respond appropriately to the stimulus.

Perception

Perception involves understanding and making sense of the world around us.

1. **Labeling**:
   - Identifying what a stimulus is.

2. **Describing:**
   - Interpreting the details about the context of the stimulus.

3. Attaching Meaning:

- Making the stimulus meaningful by relating it to past experiences and knowledge.

<u>Environmental Perception (Cognitive Bases of Perception</u>)

Environments carry meanings through their physical, social, cultural, aesthetic, and economic attributes.

1. Physical Attributes:
   - The physical setting of the stimuli.

2. Social Attributes:
   - The social context surrounding the stimulus.

3. Cultural Attributes:
   - The cultural relevance of the stimulus to the perceiver.

4. Aesthetic Attributes:
   - Personal ways of finding beauty and meaning in stimuli.

5. Economic Attributes:
- Assessing the resources required to interact with the stimulus.

Assessment of Environment

We assess the environment through:

1. **Culture**:
   - Attitudes developed from living within a culture.

2. **Needs and Preferences:**
   - Assessing according to personal needs and preferences.

3. **Future:**
   - Considering how the environment will affect us in the future.

Contextual and Social Bases of Perception

Perception is influenced by cultural, social, gender, and individual differences. These factors affect how we see and interpret our environment, showing that perception is more than just sensory input—it's embedded in a cultural context.

Lesson10
Perception and its CognitiveBases/Theories of Environmental Perception

Complex Perceptual Processes

Perceiving stimuli in the environment involves complex processes. Environmental psychologists focus on understanding how entire environments are perceived, rather than just individual stimuli. This involves the interplay of perceptual, cognitive, and evaluative processes.

Exercise

When reading Masaru Ibuka's quote, humans go through several perceptual processes:
- Start reading with a specific purpose.
- Screen out other stimuli.
- Eyes translate light waves into neural energy.

- Sensory pathways transmit electrical impulses to the brain.
- The brain organizes and interprets the input.
- The brain matches the patterns of sensation with previous images.
- Words and phrases become coherent and meaningful.

Theories of Environmental Perception

Gestalt Theory

Gestalt theory, developed by German psychologists, emphasizes the brain's active role in searching for meaning in stimuli.

The Principle of Good Form

"Gestalt" means "good form." The brain constructs meaning from stimuli, even where it might not exist. For example, many people can read a jumbled sentence as "The quick brown fox jumps over the lazy dog" due to this principle. Good form helps in understanding disorganized environments but can lead to errors in detail-oriented tasks.

The Holistic Orientation

Gestalt theory takes a holistic approach, stating that the whole is greater than the sum of its parts. For instance, we perceive a star even if there are gaps in the image. This holistic perception also applies to cognitive processes, such as remembering school days as a happy period without recalling every detail.

Size Constancy

Size constancy is the brain's tendency to interpret changes in the retinal image size as changes in distance rather than the object's size. For example, hands in a picture may appear different in size due to perceived distance, but the brain knows they are the same size.

Functionalism

Functionalism views perception as a direct process with less mediation by higher brain centers. Meaning exists in the environment, and our sensory mechanisms are prewired to respond to it. For example, we might assume a book's content by its title due to past experiences.
This approach relates to ecological biology, which studies organisms' adaptation to their environment. **For example**, animals instinctively seek environments that

offer the best survival chances. In contrast, human behavior changes constantly, making it difficult to study humans based solely on instincts.

Gibson (1979) applied this to human perception, suggesting that humans are innately able to perceive aspects of their environment that have functional value, such as infants recognizing their mother’s face for survival.

Lesson11

Learning Theory/Probabilistic Functionalism and Environmental Cognition

Learning Theory

Research on human perception emphasizes the role of experience. Perception is not innate; it develops through learning. For example, size constancy (understanding that objects remain the same size despite their distance) develops through seeing many objects at various distances. Gestalt theory and Learning perspectives both focus on the use of past experiences and direct information processing. Functionalists differ by emphasizing mechanical, instinctual behavior without involving higher brain centers.

Learning theorists suggest that experience leads to assumptions about the world. For instance, youngsters might assume smoking makes them look stylish due to movie heroes, leading them to smoke despite its harmful effects. These assumptions save time and effort in new situations by drawing from past experiences. However, assumptions can also lead to misperceptions or illusions, as demonstrated by optical illusions like the Ames room.

Probabilistic Functionalism

Given the complexity of environments, understanding how entire environments are perceived is challenging. Brunswick’s lens model (1956, 1969) is a key learning theory in environmental psychology, offering a way to mathematically describe perceptual processes in complex environments.

- **Ecological Validity:** The relationship between specific stimuli and the environment, focusing on relevant information.

- **Functional Validity:** The usefulness of the information our perceptual lens focuses on, validated through actions.

Brunswick's model suggests that we filter environmental stimuli through a perceptual lens, emphasizing important cues and ignoring others. This helps simplify complex environments and avoid misleading sensory information.

Environmental Cognition

Difference between Perception and Cognition

- **Perception**:
Gathering sensory information about the environment (e.g., seeing a rose and noticing its color and fragrance).

- **Cognition**:
Processing and interpreting this sensory information (e.g., thinking about the symbolic meanings of a rose).

Environmental cognition involves thinking about the environment, processing information, organizing knowledge, and understanding how this knowledge is acquired and varies among individuals.

Responses to Novel Environments

When encountering a new environment, people typically experience several responses:

- **Affect**: Emotional response to the new environment, such as feeling happy, secure, sad, or insecure. First impressions can have lasting impacts.
- **Orientation**: Actively seeking to understand and find one's place in the new setting, a cognitive process involving exploration.

These responses help individuals adapt to and feel in control of their new surroundings. The six responses to novel environments (affect, orientation, categorization, systemization, manipulation, and encoding) will be discussed in detail in the next lecture.

Lesson #12
Responses to Novel Environments and Environmental Cognition

Responses to Novel Environments

A novel environment is an unfamiliar setting encountered for the first time or after a long absence. According to Ittleson, Proshansky, Rivlin, & Winkel (1974), six types of responses occur: affect, orientation, categorization, systemization, manipulation, and encoding. These responses often overlap.

Affect:

Our emotional response to new environments varies. We feel heightened awareness and arousal due to the need to understand and feel secure. Specific settings can evoke happiness or fear, influencing our future interactions with the environment.

Orientation:

In a new setting, we try to find our place. This cognitive process involves locating important places like the dining hall, library, or where to buy books. It's about knowing where we are in relation to the environment.

Categorization:

Beyond orienting, we evaluate and give meaning to the environment based on our needs and values. For example, we look for the best or cheapest pizza. Categorization helps us understand the environment better by distinguishing between various options.

Systemization:

We organize the environment into meaningful structures, such as knowing the best time to visit the library for parking or finding helpful staff. This involves creating more complex understandings of our surroundings.

Manipulation:

With systemization, we achieve a sense of order and can predict and control the environment to our advantage. For example, if a preferred restaurant is closed, we know alternative options.

Encoding

Encoding is the highest level of understanding, allowing us to use shared symbols and navigate the environment more effectively. It enables us to “think-travel” through environments and adapt to changes.

Characteristics of Cognitive Maps

A cognitive map is a mental representation of the physical environment, helping us navigate and interact with our surroundings. Lynch (1960) identified five major characteristics from studying city residents’ maps:

Paths

Major routes of traffic flow, like Main Street.

Edges

Lines that divide areas or mark boundaries, such as a river.

Districts

Large sections with distinct identities, like Chinatown.

Nodes

Points where major routes intersect, such as a busy intersection.

Landmarks

Unique structures used as reference points, like a tall building.

Lesson #13
Characteristics of Cognitive Maps

A common method for studying spatial cognition is asking people to draw “sketch maps” of environments. Sketch maps are reliable data collection tools (Blades, 1990). Lynch (1960) studied residents of Boston, Los Angeles, and Jersey City by having them draw maps of their cities. He identified five key features of cognitive maps:

1. ****Paths****: Major traffic routes (e.g., Main Street).
2. ****Edges****: Lines that divide or bound areas (e.g., a river).
3. ****Districts****: Large sections with distinct identities (e.g., Chinatown).
4. ****Nodes****: Intersections of major routes (e.g., a busy street corner).
5. ****Landmarks****: Unique structures used as reference points (e.g., a tall building).

<u>These elements create a "cognitive space" in our minds, organized into distinct regions.</u>

Anchor-Point Hypothesis

This idea suggests that prominent cues, like key landmarks or nodes, help organize cognitive space. These anchor points provide the framework for our mental maps.

Factors Affecting Cognitive Maps

****Environmental Differences****

Cities differ in how easily people can form cognitive maps. Lynch (1960) used the term "legibility" to describe how well a city's layout is understood by its inhabitants. Recognizable areas help in developing accurate maps, as environments with prominent, frequently passed structures make it easier to form clear mental images.

Errors in Cognitive Maps

Cognitive maps often contain errors due to human spatial cognition limits. Common errors include incomplete maps, distorted distances, and inclusion of non-existent elements. Personal significance can also distort maps, such as drawing one's own country larger in world maps.

Individual Differences

Some people are better at forming cognitive maps than others. Men often create more accurate maps than women, using more spatial terms and fewer errors. Socioeconomic status also affects map accuracy, with higher status individuals having more complete maps. Familiarity with an area improves map accuracy.

Familiarity and Experience

Experience and familiarity with an environment are key in forming accurate cognitive maps. Mobility and opportunities for travel play significant roles. For instance, those who travel more often have better mental maps of larger areas.

In conclusion, while individual spatial ability matters, experience and familiarity with the environment are crucial for accurate cognitive maps.

lesson #14
Environmental Evaluation

Understanding how we evaluate environments is complex but essential for predicting behavior. Generally, people prefer and move towards environments they like, and avoid those they don't. Evaluations can be based on preferences, judgments of beauty, or positive reactions. Different approaches identify various aspects of the physical environment that influence these evaluations.

Kaplan's Model of Environmental Preference

Kaplan (1975) developed a model to predict why some environments are preferred over others, based on their informational content. Preferences depend on:

1. ****Coherence**:** How organized and harmonious the environment is, with clear patterns and relationships.
2. ****Legibility**:** How easy the environment is to understand and navigate, with distinctive features and landmarks.
3. ****Complexity**:** The variety and diversity of elements, colors, shapes, and textures, without being overwhelming.
4. ****Mystery**:** The extent to which the environment invites exploration and discovery, with hidden or partially revealed elements that arouse curiosity.

People prefer environments high in coherence and legibility, and also appreciate complexity and mystery.

Biological Basis to Preferences

Kaplan argues there is a survival value in preferring environments that provide informational advantages. Research highlights the importance of mystery in landscape preferences but also applies to urban and interior environments.

Environmental Aesthetics

Aesthetics is our sense of appreciating beauty. While beauty can be subjective, psychologists have identified common features in environments considered beautiful. For example, most people find the Grand Canyon beautiful and a junkyard ugly. By studying various settings, we can identify what makes an environment aesthetically appealing.

Environmental Aesthetics will be discussed in detail in the next lesson.

Lesson #15:
Environmental Evaluation Affective Bases

Environmental Aesthetics

Psychologists aim to identify objective dimensions that lead to judgments of aesthetic appeal. Berlyne (1960) identified four collative properties:

1. **Complexity**: The variety of elements in the environment.
2. **Novelty**: The presence of stimuli not previously encountered.
3. **Incongruity**: The extent to which stimuli seem mismatched.
4. **Surprisingness**: The presence of unexpected elements.

Affective Bases for Environmental Evaluation

- **Wohlwill's Butterfly Curve Hypothesis:** Predicts a curvilinear relationship between positive affect and deviations from the adaptation level. Moderate deviations are pleasant, extreme deviations lead to negative affect.

- **Mehrabian & Russell's Model**: They introduced the concept of information rate to define environmental stimulation level. Arousal correlates with information rate. Approach behaviors are highest at intermediate levels of arousal, moderated by the degree of pleasure.

In the next lecture, we'll explore the remaining aspects of Affective Bases for Environmental Evaluation.

Lesson #16:
Environmental Attitudes**

Russell & Pratt Model of Emotional Reactions to Environment
Russell & Pratt (1980) proposed a model with pleasure and arousal as independent dimensions, creating a circular ordering of emotional quality in environments. It distinguishes between pleasant and unpleasant environments at varying arousal levels.

Environmental Attitudes
Attitudes consist of cognitive, affective, and behavioral components:
- **Cognitive Component**: Involves perception and understanding of the environment.
- **Affective Component**: Refers to emotional reactions towards the environment.
- Behavioral Component: Reflects actions influenced by cognition and affect.

Environmental Attitude Formation
- **Beliefs**: Form the cognitive "building blocks" of attitudes. Divided into primitive and higher-order beliefs.
  - **Primitive Beliefs:** Accepted as givens, often based on direct experience or external authority.
  - **Higher-order Beliefs**: Involve conscious premises in arriving at beliefs, like understanding air pollution's impact on health.

In the next lecture, we'll delve deeper into types of beliefs and their role in shaping environmental attitudes.

Lesson # 17
EnvironmentalAttitudeFormation

Environmental Attitude Formation

Depth and Breadth of Beliefs:
- **Depth:** Multiple premises leading to the same conclusion.

- **Breadth**: Many syllogisms or deductive reasonings leading to the same conclusion.
- Higher-order beliefs can be irrational yet internally consistent, influencing attitudes and evaluations.
- **Values**: Basic preferences for certain end states. Values guide attitudes towards environmental concerns.

Affective Bases for Attitude
- The affective component of an attitude is often learned through classical conditioning.
- Byrne-Clore reinforcement-affect model explains how neutral stimuli acquire the ability to elicit responses through association.

- Example: Associating a neutral stimulus (like a desk) with a negative stimulus (like loud music) can lead to a negative affective response towards the desk.

Behavioral Bases of Attitudes

- Operant Conditioning: Behaviors followed by pleasant consequences increase in frequency. Behaviors followed by unpleasant consequences are extinguished.
- Attitudes, being a form of behavior, can be operantly conditioned. We learn to express attitudes that lead to favorable outcomes.

- **EXAMPLE**: Pigeons were conditioned in Skinner's experiments using a specially designed box called the Skinner Box.

Next Lecture: Instrumental Conditioning
- Instrumental conditioning will be discussed in the next lecture.

Lesson # 18
SocialBasesof Attitudes

Social Bases of Attitudes:
Operant Conditioning (Skinner, 1938)
- Describes how behaviors are influenced by their consequences.

- **Reinforcement:**
  - Positive Reinforcement: Adding a desirable stimulus to increase behavior.
  - **Negative Reinforcement**: Removing an aversive stimulus to increase behavior.
- **Punishment**:
  - **Positive Punishment**:Adding an aversive stimulus to decrease behavior.
  - **Negative Punishment**: Removing a desirable stimulus to decrease behavior.

Social Learning

- **Observation of others' behavior**: influences attitude formation.
- Social Learning: Learning by observing a model's behavior.
- Vicarious Learning : Influenced by the consequences of the model's behavior.

Impact of Social Influences on Attitudes

- People imitate attitudes that lead to favorable consequences.
- Social dynamics, such as friendships, can reinforce attitudes.
- **Attitude-Behavior Consistency:** People strive to maintain logical consistency between their attitudes and behaviors.
- **Cognitive Discomfort:** Mental stress when holding contradictory beliefs or attitudes, motivating individuals to restore consistency.

Example:

- If someone opposes pollution control but realizes the inconsistency with their belief in environmental protection, they might change their attitude or behavior to restore consistency.

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LESSONS 1-18
THE END